

LiDAR Crime Scenes for VR

Project Plan

**COMP3018 - Professional Experience
Western Sydney University**

Date: 21/03/2025

Group: PA2520

Supervisor: Marvis Leung

Team Members:

- Zhanhao Chen
- Ujwal Mainali
- Yelin Naung
- Yunpeng Wang

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Executive Summary

This project expands on the original Microsoft HoloLens 2 holographic crime scene application to operate on the Meta Quest 3. The new application gives forensic investigators the ability to engage with LiDAR-scanned crime scenes by using the depth sensing and pass-through features and networking capabilities of Quest 3 within a shared mixed reality framework.

The project execution follows a structure which begins with research and planning phases and proceeds through system design followed by development and testing then final handover. A Gantt chart included in the appendix presents the project timeline with key phases beginning from proposal submission through feature development to final testing.

The final project includes a tested application using Proton SDK for multi-user interface along with documentation and a handover report. The project focuses development on technical feasibility, usability and performance standards to satisfy client and user expectations.

Introduction

Project planning is an important part of our team's systematic implementation of the LiDAR Crime Scenes for VR project. The team mainly adopts a structured planning approach to assign team tasks, covering detailed task decomposition, potential risk identification and time mitigation strategies, and team project constraints. This allocation is to ensure the smooth progress of the project.

WBS (Work Breakdown Structure) is to break down the project into specific tasks for each part, thereby helping team members better allocate resources and manage work. We recorded the potential risks of the project and specified mitigation plans for the problems. In addition, we also clearly annotated the project scope constraints to increase the feasibility of the project while it is ongoing, shorten the response time, and leave more buffer time and space.

We also created project milestones to ensure that the project can track progress accurately and effectively. Personal task lists and communication plans allow each team member to have a clearer understanding of the project's mastery and clarify their personal positions. This can closely connect the team and stakeholders and maintain consistency. Finally, based on the previous content, we use a detailed Gantt chart to integrate all elements. The Gantt chart shows the timing and coordination of each task. In general, the purpose of project planning is to comprehensively plan VR development projects and improve response capabilities and development comfort throughout the entire project development cycle.

Group Details

Name	Relevant Skills	Strength	Weakness
Yelin Naung	Main coding language is C++	Experienced coding with C++, transferable skill when using C#	Lack of experience with unity development
Zhanhao Chen	Unity operating	Experienced Unity, extra flexible in C++ coding environment	Lack of experience with Coding
Yunpeng Wang	Front-end and mobile app development	Experienced coding with C#	Lack of experience with unity development
Ujwal Mainali	Familiar with C++, javascript	Working in group, able to adapt with new tools and framework	Lack of knowledge on Unity

Collective Analysis

The team brings together members with skills in C++ and C#, along with experience in Unity, front-end development, and adaptability. However there are gaps in Unity Development and VR-specific implementation, which needs to be addressed for the successful completion of the project.

Strengths:

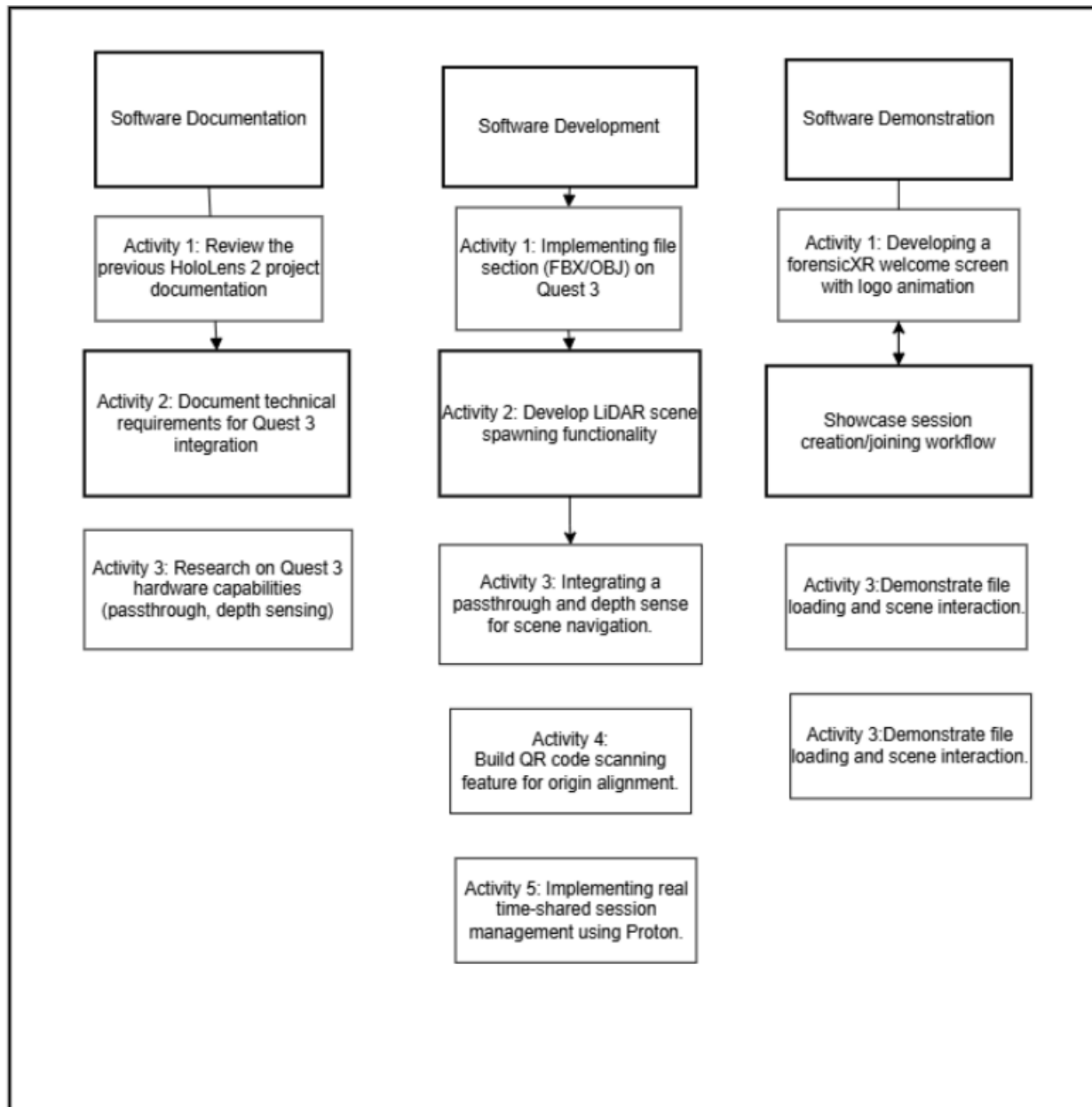
- Background in C++: as majority of the team has experience in C++, which provides a transferable skill for C# programming in Unity.
- We also have a member with experience in Unity, making them well suited for guiding development and implementing VR functionalities.
- Having a member with experience in front-end and mobile development will be useful for designing and implementing the UI/UX components with the VR environment.

Challenges and Mitigation:

- Limited Unity Development Experience
 - Conduct team workshops or self-paced learning using available resources on the internet.
- Minimal Experience with VR-Specific Features
 - Allocate time for research and testing of all Meta Quest 3 features. Reviewing Previous project work will also provide a reference for implementation.

To conclude, while the team has strong foundation in C++ and C#, the gaps in Unity and VR development will require learning and collaboration. By using the existing strengths and allocating time for learning, the team can have a smooth and efficient development process.

Work Breakdown Structure (WBS)



Project Related Risks

	Risk	Mitigation Strategies	Impact level
1	Missing critical project deadlines due to scheduling conflicts	Set regular team contacts, anticipate and adjust personal schedules at any time, and effectively use project buffer time	Medium
2	Delivery delays due to sick team members	Ensure that when one team member is sick, other members can take over the follow-up development	Low
3	Unclear goals and misunderstandings	Update development progress in real time and conduct regular coordination tasks to ensure that target requirements are correctly understood	Medium
4	Key file or document storage issues	Use a cloud backup strategy to ensure that it is always turned on during personal development to avoid software and device problems that may cause development progress to be lost (e.g. GitHub, Google Drive)	Medium
5	Lack of experience in VR and LiDar application development	Make sure to take online training courses before starting development, or seek help from your supervisor	High
6	Poor communication leads to team collaboration problems	Ensure efficient use of multiple hybrid communication tools for communication and development progress sharing	Medium
7	Lack of hardware VR equipment (Quest 3)	Need to arrange for use of alternative equipment or technical support resources	High
8	Project deliverables do not meet customer expectations	Establish clear acceptance criteria early; demonstrate progress regularly and collect feedback	High

9	Insufficient testing or progress verification	Customised test plans and frequent coordinated development verification to ensure that there are no remaining issues in the content jointly developed by members	High
10	potential communication issues between stakeholders	In the early stages of the process, have regular meetings, clearly define the agenda, prepare for each meeting with client or supervisor, and update the records afterwards	Medium
11	Disengaged or unmotivated team members	Establish clear and fair task allocation, report any problems to other members immediately, and hold regular meetings for feedback	Low
12	Infrequent updates lead to customer dissatisfaction	Provide progress reports at every routine client meeting	High
13	VR motion sickness	It can be relatively adjusted during the development period to make the wearing experience more comfortable.	Low

Project Related Constraints

	Constraint	Mitigation Strategies	Type
1	Project must be completed within 15 weeks (semester length)	Strictly follow a detailed timeline and track milestones weekly via Gantt charts	Time
2	IP transfer required before and after project completion	Prepare and clearly document all project-related intellectual property and assets before project development	Legal
3	Team experience constraints	Assign respective tasks at the beginning of the project and train yourself within a manageable time to adapt to development requirements	Scope
4	Client's project documentation and system clarity requirements	Verify document quality regularly and record detailed meeting logs	Quality
5	Quest 3 hardware/software requirements	Search for available equipment and verify hardware as early as possible, prepare for development testing, and arrange alternative tools or virtual machines if necessary, Configure access rights to the LiDAR device in advance	Resources
6	Compatibility constraints between VR development software versions	Ensure the same version of development software between teams	Scope
7	Must use Quest 3 SDK	Schedule additional time for familiarisation SDK	Resources
8	Collaborative team meetings limited by physical workspace availability	Because VR development may require multiple interactive tests, it is necessary to arrange virtual or face-to-face meetings and clarify the requirements of each meeting	Resources

9	Stakeholder schedule conflicts	Arrange scheduled meetings in advance, prepare briefings and any development issues, record and plan during the meeting, and prepare for subsequent development, Always adhere to professional ethics in each meeting and report project progress regularly	Time
10	Use of university labs restrictions	Confirm laboratory opening hours, ensure security licences and comply with terms of use, and arrange remote work alternatives if necessary	Resources
11	Semester break and holiday time constraints	Ensure that more buffer time is provided for the project during the holidays to increase development flexibility	Time

Project Milestone

Milestone	Responsibility	Planned Date
Information and Resource Gathering	Group	14/03/2025
Project Planning Finalised	Group	21/03/2025
Project Proposal	Group	28/03/2025
Initial Prototype Development	Group	11/04/2025
System Analysis and Design Document	Group	18/04/2025
Development of Final System	Group	30/05/2025
Handover and Completion Report	Group	30/05/2025

Individual Tasks List

Team Member: Zhanhao Chen				
<u>Task</u>	<u>Deliverable</u>	<u>Start Date</u>	<u>End Date</u>	<u>% Complete (as of reporting)</u>
Project Related Risks	Project Plan	17/03/2025	21/03/2025	100%
Project Related Constraints	Project Plan	17/03/2025	21/03/2025	100%
Individual Task	Project Plan	17/03/2025	21/03/2025	100%
High-Level Business Functions	Project Proposal	24/03/2025	28/03/2025	0%
Feature/Function Test Plan	System Analysis	14/04/2025	18/04/2025	0%
Screen Designs	System Analysis	14/04/2025	18/04/2025	0%
Use Case Diagram	System Analysis	14/04/2025	18/04/2025	0%
List of Messages	System Analysis	14/04/2025	18/04/2025	0%
Candidate Test List	Handover Completion Report	26/05/2025	30/05/2025	0%

Team Member: Yelin Naung				
<u>Task</u>	<u>Deliverable</u>	<u>Start Date</u>	<u>End Date</u>	<u>% Complete (as of reporting)</u>
Group Milestone Plan	Project Plan	17/03/2025	21/03/2025	100%
Gantt Chart	Project Plan	17/03/2025	21/03/2025	100%
Individual Task	Project Plan	17/03/2025	21/03/2025	100%
Functional Requirements	Project Proposal	24/03/2025	28/03/2025	0%
Sequence Diagram	System Analysis	14/04/2025	18/04/2025	0%
Entity Relationship Diagram	System Analysis	14/04/2025	18/04/2025	0%
Database Schema	System Analysis	14/04/2025	18/04/2025	0%
Functional Variations	Handover Completion Report	26/05/2025	30/05/2025	0%

Team Member: Yunpeng Wang				
<u>Task</u>	<u>Deliverable</u>	<u>Start Date</u>	<u>End Date</u>	<u>% Complete (as of reporting)</u>
Communication Plan	Project Plan	17/03/2025	21/03/2025	100%
Individual Task	Project Plan	17/03/2025	21/03/2025	100%
Cost Benefit Analysis	Project Proposal	24/03/2025	28/03/2025	0%
Risks and Issues Related to the System	System Analysis	14/04/2025	18/04/2025	0%
Constraints Related to the System	System Analysis	14/04/2025	18/04/2025	0%
Software Architecture Diagram	System Analysis	14/04/2025	18/04/2025	0%
Data variation	Handover Completion Report	26/05/2025	30/05/2025	0%

Team Member: Ujwal Mainali				
<u>Task</u>	<u>Deliverable</u>	<u>Start Date</u>	<u>End Date</u>	<u>% Complete (as of reporting)</u>
Work Breakdown Structure (WBS)	Project Plan	15/03/2025	21/03/2025	100%
Individual Task List	Project Plan	17/03/2025	21/03/2025	100%
Network and Communication Diagram	System Analysis	14/04/2025	18/04/2025	0%
Expanded Use Cases	System Analysis	14/04/2025	18/04/2025	0%
Candidate Tests	System Analysis	14/04/2025	18/04/2025	0%
Screen Variations	Handover Completion Report	26/05/2025	30/05/2025	0%
Conclusion	Handover Completion Report	26/05/2025	30/05/2025	0%

Communication Plan

Meeting Type	Length	First Meeting	How Often	Number of Meetings Planned	Location/Mode
Industry Partner Meetings	60 mins	14/3/2025	Every two weeks Friday	7	Online/Zoom
Academic Supervisor Meetings	60 mins	11/3/2025	Weekly Tuesday	14	Online/Zoom
Formal Team Meetings	60 mins	10/3/2025	Weekly Monday	14	Online/Zoom or Discord
Casual Team Meetings	30 mins	08/03/2025	Weekly when needed	28	Online/Zoom or WhatsApp

Gantt Chart

Project timeline becomes visible through Gantt charts because this tool displays each phase with detailed tasks that depend on one another. The Gantt chart serves to maintain process tracking and designated task assignments and makes project execution streamlined.

This Gantt chart includes:

- **Project Planning and Research:** Initial meetings, document gathering, and defining milestones.
- **Proposal Phase:** Scope definition, functional and non-functional requirements, and Cost-Benefit Analysis.
- **System Analysis and Design:** Software architecture, UI/UX planning, and database design.
- **Development Phase:** Divided into core features, network functionality (Proton/QR), and Quest 3 depth sensing/pass-through integration.
- **Testing and Debugging:** A structured approach to ensure all components function correctly before finalisation.
- **Finalisation and Handover:** Document completion, client meetings, and submission of the Handover and Completion Report.

Each phase is colour-coded to enhance readability and highlight task dependencies.

The complete breakdown chart exists as a distinct PDF in the appendix section.

Conclusion

This project aims to extend the capabilities of the existing holographic Crime Scene application from HoloLens 2 to Meta Quest 3, leveraging Unity MR, Meta XR SDKs, and Proton for multiplayer functionality. This project will implement features on Quest 3, including file selection (FBX/OBJ), scene spawning, movement and rescaling, QR code-based origin setting, and pass-through with depth sensing. By overcoming technical challenges in compatibility, interaction mechanics, and network synchronisation, this project will push the boundaries of crime scene visualisation and analysis.

Appendix

1. Gantt Chart

The Gantt chart provides a detailed breakdown of the project timeline, including task dependencies, milestones, and assigned responsibilities. It is included as a separate PDF file for clarity and readability.

Reference

Project Documents

Western Sydney University 2025, COMP3018 Professional Experience – Assessment Guide, Western Sydney University, viewed 21 March 2025, available upon request.

Western Sydney University 2025, Project Brief – PA2520: LiDAR Crime Scenes for VR, Western Sydney University, viewed 21 March 2025, available upon request.

Other Sources

Smartsheet 2025, 'Guide to Software Project Management', Smartsheet, viewed 21 March 2025, <https://www.smartsheet.com/content/software-project-management>.

TeamGantt 2025, 'Software Development Project Plan Template', TeamGantt, viewed 20 March 2025, <https://www.teamgantt.com/software-development-templates/software-development-plan>.